

1 Physical System Setup & SITL Architecture

The autonomous mission adaptation system integrates physical camera hardware, software-based perception, high-level decision-making, and Software-in-the-Loop (SITL) flight simulation. The system architecture consists of four main components:

- **Physical Optical Sensing:** A smartphone camera, connected via the IP Webcam application over a shared Wi-Fi network, serves as the live camera feed. The phone streams video over HTTP, which is captured on the companion workstation using OpenCV's `VideoCapture` interface.
- **Companion Workstation:** A laptop running Ubuntu 24.04 with ROS 2 Jazzy processes the live video feed. A dedicated ROS 2 node (`camera_publisher_node`) captures frames from the phone stream and publishes them on the `/camera/image_raw` topic using the `sensor_msgs/Image` message type via `cv_bridge`.
- **ROS 2 / MAVROS Middleware:** A second node (`vision_processor_node`) subscribes to the image stream, performs the perception and decision logic, and communicates with MAVROS through its service interface (e.g., `/mavros/set_mode`) to convert high-level decisions into MAVLink commands.
- **Flight Execution & Telemetry:** The MAVLink commands are received by a simulated ArduPlane instance running inside ArduPilot SITL. Flight mode changes and telemetry are monitored live in QGroundControl (QGC).

Figure 1 shows the system running live: the camera feed window (*PADA Live Vision & Execution Feed*) displays the phone stream, the terminal log confirms that `vision_processor_node` detected a valid frame and issued a mode-change command, and QGroundControl confirms the resulting flight mode change to **GUIDED** in real time. This demonstrates a working end-to-end pipeline connecting the physical camera, the ROS 2 nodes, and the SITL-simulated aircraft.

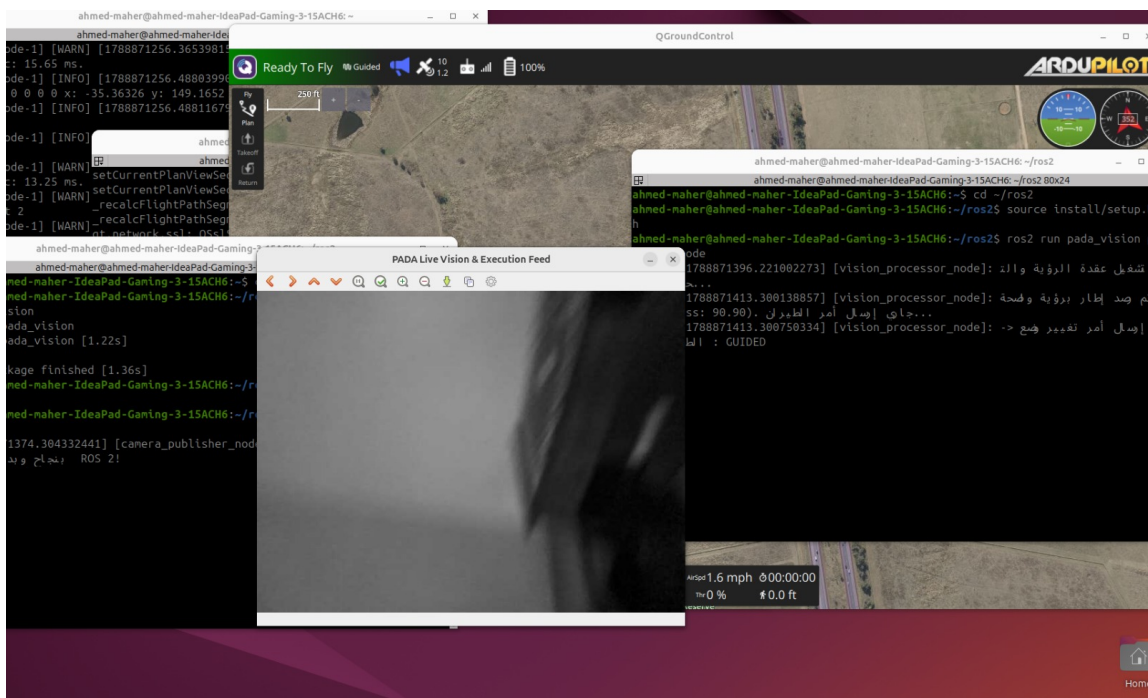


Figure 1: Live system operation showing the camera feed window, the ROS 2 node terminal logging the detection and mode-change command, and QGroundControl confirming the resulting GUIDED mode.